

## 1.11 Test and Evaluate HCI (Evaluation Methods)

ٹیسٹنگ یہ دیکھتی ہے کہ پروڈکٹ ٹھیک چلتا ہے (بگز)؛ ایویلیوایشن یہ دیکھتی ہے کہ صارف کی ضرورت پوری ہوئی۔

CS XII

Chapter 1: Computer Systems (HCI)

Lecture 14 of 16

## 1.11 Test and Evaluate HCI (Evaluation Methods)

### LEARN IT

**Testing** checks if the product works properly (finding bugs). **Evaluation** determines whether it meets users' needs (usability). Together they improve the product before release.

A school portal can be "bug-free" (testing passed) and still unusable if students cannot find the date sheet (evaluation failed). You need both.

### 1.11.1 Evaluation methods

| Method                       | Description                                                                                   |
|------------------------------|-----------------------------------------------------------------------------------------------|
| SUS (System Usability Scale) | A standard 10-item questionnaire used to calculate a score for how usable a system is.        |
| NPS (Net Promoter Score)     | Asks users: "How likely are you to recommend this app to a friend?" Measures user loyalty.    |
| Formative evaluation         | Done <b>during</b> development to collect early feedback and improve the design continuously. |
| Summative evaluation         | Done at the <b>end</b> of development to measure overall success against the original goals.  |
| Expert reviews               | Usability specialists review the app against established design standards (heuristics).       |

**Heuristics** are agreed rules of good design (for example "show system status", "be consistent"). An expert review checks the product against those rules without waiting for a huge user survey.

### WORKED EXAMPLE

**Worked example: when to use each method on a school portal.**

1. Week 2 of building: show the Lo-Fi to five students, change the menu. That is **formative** evaluation.
2. A teacher who has studied HCI checks the screens against standard rules (Is feedback visible? Are labels consistent?). That is an **expert review**.
3. After the portal is "finished", 30 students fill the 10 SUS questions. A low score means it is still hard to use, even if there are no crashes.
4. Ask: "On a scale of 0–10, would you recommend this portal to a friend in another college?" That is **NPS** (loyalty).
5. Compare the finished portal with the original goals (date sheet in one tap, bilingual fee notice). That is **summative** evaluation.

### REMEMBER IT

**Formative = forming it (during).** **Summative = summing it up (end).** SUS = 10 questions on usability. NPS = "Would you recommend it?" Expert = specialist vs heuristics.

### COMMON MISTAKE

**Mistake:** Saying testing and evaluation are the same, or mixing formative with summative.

**Fix:** Testing = bugs. Evaluation = does it meet human needs. Formative is during; summative is at the end against original goals.

### اردو میں سمجھیں

ٹیسٹنگ یہ دیکھتی ہے کہ پروڈکٹ ٹھیک چلتا ہے (بگز)؛ ایویلیویشن یہ دیکھتی ہے کہ صارف کی ضرورت پوری ہوئی۔ SUS دس سوالوں والا استعمال کا اسکور ہے، NPS پوچھتا ہے کہ آپ یہ ایپ دوست کو کتنی تجویز کریں گے۔ فارمیٹو جانچ ترقی کے دوران ہوتی ہے، سمیٹو آخر میں اہداف کے خلاف۔ ماہر جائزہ ہیورسٹکس (Heuristics) پر ہوتا ہے۔

### CHECK YOURSELF

1. What is the difference between testing and evaluation?
2. How many items are on the SUS questionnaire?
3. When is summative evaluation done?

Answers: 1. Testing finds bugs / checks the product works; evaluation checks it meets users' needs (usability). 2. 10. 3. At the end of development, against the original goals.

---

Next lecture: 1.11.2 Testing Methods (Usability, A/B, Automated)